Owen Ritchie

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Data-driven engineer with 2+ years of experience building cloud data pipelines, dashboards and ML models across SaaS, government and consulting. Graduating Dec 2026, seeking full-time data, analytics or ML roles; planning part-time graduate study in data science.

Education

Honours Bachelor of Science, Computer Science — *University of Ottawa* **Sep 2022–Dec 2026**

- Dean's List. Relevant coursework: Modelling & Simulation, Artificial Intelligence, Design & Analysis of Algorithms, Cybersecurity

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Work Experience

Data & Analytics Engineer Intern — *Kinaxis* **Sep 2025–Aug 2026**

- Architected ELT pipelines for **3** Google Analytics properties, processing **250k+** daily events in GCP with dbt, GCS & Cloud Run
- Cut **100k+** CU-hours of yearly Fabric compute (**~$18k saved**) by optimizing Power BI dataflows with BigQuery, dbt & DAX
- Migrated **~90M** rows of Eloqua campaign data from Snowflake to BigQuery with zero downtime using dbt & Snowsight
- Created a BigQuery ML classifier that labels Salesforce opportunities as Win/Loss from SDR comments, replacing manual review
- Developed a CEO-facing customer footprint tool aggregating **500M+** concurrent tracked parts, scenarios & users across **400+** customers, using an internal log collector API, dbt, BigQuery & Datadog
- Built a Chrome extension to optimize GCP pipeline cost against committed cloud spend, using Vertex AI, TypeScript & BigQuery
- Centralized **10+** years of ServiceNow & LogicMonitor data in GCS with dbt, saving the FinOps Lead **2+** days of monthly reporting

Data Analytics Intern — *Global Affairs Canada* **Jan 2025–Jul 2025**

- Validated **5+** years of cross-institutional data backing **50+** statistics in Canada's G7 Accountability Report, using Python
- Reduced API response times by **99.8%** by implementing multithreading to ingest data from the World Bank, OECD, UN and WEF
- Deployed a development indicators tool to **10+** Directors to model the impact of policy decisions across aid funding scenarios
- Delivered reports on the Sahel and CARICOM covering **25+** metrics, including HDI and electrification, using Matplotlib, Pandas & Scikit-learn to estimate missing indicator values

Data Science Intern — *Poisson Consulting Ltd.* **May 2023–Sep 2023**

- Analyzed **500k+** salmon egg records in R & Tidyverse, supporting publications led by senior computational biologists
- Updated R analysis scripts for compatibility with current package releases, keeping published analyses reproducible every year
- Mapped **3k+** salmon redds from weekly drone surveys in QGIS to estimate how dam operations affected egg survival
- Volunteered as a TA at Columbia Mountains Institute of Applied Ecology, teaching **15** students to apply R in ecology

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Personal Projects

owenritchie/data — *dbt, SQL, Python, GitHub Actions, Neon (Postgres)* **2025–Present**

- Built an analytics repository that automates ingestion (Python) and transformation (dbt) for my personal projects
- Powers *Spotify Analytics* (React, my listening history) and *OC Transpo Analytics* (Streamlit, Ottawa bus fleet monitoring)

Mill — *Next.js, React, TypeScript, Tauri, Rust, Drizzle, Claude SDK* **2026–Present**

- Built a full-stack macOS project planner with a node-based canvas and chat-based planning via Ollama & Claude models
- Packaged the app as a .dmg with Tauri & a GitHub Actions release workflow, storing data locally in libSQL via Drizzle

CSI 4900 Project Portal — *TypeScript, SQL, Supabase, AWS S3, Terraform* **2026–Present**

- Designed the Supabase Postgres schema for a student-professor project portal used by **200+** students each semester
- Implemented row-level security and a custom JWT hook enforcing user roles across projects, inquiries and attachments
- Wrote **320+** pgTAP tests covering schema, RLS policies and the authentication hook

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Technical Skills

Languages: Python, SQL, R, TypeScript, JavaScript, Java, Bash

Frameworks & Libraries: dbt, Pandas, Scikit-learn, Apache Spark, TensorFlow, OpenAI, Vertex AI, Claude SDK, React, Node.js

Infrastructure: BigQuery, Snowflake, Postgres, GCS, AWS S3, Cloud Run, GitHub Actions, Airflow, Supabase, Linux, Docker, Git, REST APIs

Reporting: Power BI, Looker Studio, Grafana, Datadog, Metabase, R Markdown, Jupyter, Google Analytics

AI-Development Tools: Claude Code, Devin, Codex, Google Antigravity (Gemini), GitHub Copilot

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